

CHAPTER 12 TEMPERATURE - NOTES

12.1 THERMAL EQUILIBRIUM

1. Temperature tells us about the hotness and coldness of a body with respect to some standard.

2. Temperature is average kinetic energy of molecules in a matter.

3. What is meant by temperature?

Physical properties of material (body) to determine in which direction heat is transferred from one object to another due to a temperature differences between the two objects.

4. A thermometer is a common instrument used to measure temperature.

5. Two bodies in contact with each other are said to be in thermal equilibrium when the two bodies have the same temperature.

6. When two bodies are not in same temperature, are put in contact, then thermal energy (heat energy) flows from the hotter to the cooler body until thermal equilibrium is established.

7. Ex.

8. During the process of reaching thermal equilibrium between two bodies, there will be net thermal energy flow.

9. The hotter body loses internal energy while the cooler body gains in internal energy.

Recall internal energy:

12.2 THERMOMETRIC SCALES

1. Various degree of hotness is represented by a temperature scale.
2. In order to establish a thermometric scale, we need:
 - i) a thermometric property i.e. a physical property that varies consistently with degree of hotness.
 - ii) Two fixed point i.e. two degree of hotness which is easily and accurately reproducible.
3. Example of thermometric properties are:
 - a) volume of a liquid or length of a liquid column.
 - b) pressure of a fixed mass of gas at constant volume.
 - c) resistance of a coil of wire.
 - d) e.m.f. of a thermocouple.
4. There are many fixed points in nature. The two points most often used are:
 - i) the ice-point, which is the temperature of pure melting ice, assigned a value of 0°C .
 - ii) the steam point which is the temperature of the steam from pure water boiling under standard pressure, assigned a value of 100°C .
5. In order to establish a temperature scale:
 - i) measure x_0 , the chosen thermometric property at the ice-point.
 - ii) measure x_{100} , the chosen thermometric property at the steam point.
 - iii) measure x_{θ} , the thermometric property at the unknown temperature $\theta/^{\circ}\text{C}$.
 - iv) calculate θ from the equation:

$$\theta/^{\circ}\text{C} = \left(\frac{x_{\theta} - x_0}{x_{100} - x_0} \right) 100$$

6. These equation and its rationale can be illustrated graphically :

12.3 TYPES OF THERMOMETER

1. Liquid-in-Glass thermometer
2. Constant volume gas thermometer
3. Platinum Resistance thermometer
4. thermoelectric thermometer

Comparison of Thermometers

4. A COMPARISON OF THE THERMOMETERS

NAME OF THERMOMETER	MAXIMUM TEMPERATURE RANGE (°C)	ADVANTAGES	DISADVANTAGES
(a) mercury-in-glass	-39 to 500	(i) portable (ii) direct reading	(i) relatively small (ii) not very accurate
(b) constant volume gas	-270 to 1500	(i) very wide range (ii) very accurate (iii) very sensitive	(i) cumbersome (the bulb may have a capacity of about 10^{-3} m^3) (ii) very inconvenient and slow to use
(c) platinum resistance	-200 to 1200	(i) wide range (ii) best for small, steady differences of temperature (iii) the most accurate thermometer in the range -200 °C to 600 °C	not suitable for varying temperatures because of (i) large heat capacity, and (ii) use of materials of low thermal conductivity in the sheath
(d) thermocouple	-250 to 1500	(i) wide range (ii) very small, so it is useful for both local and rapidly varying temperatures (iii) the most accurate thermometer in the range 600 °C to 1065 °C	not as accurate as (c) in the range -200 °C to 600 °C